

[UNCONVENTIONAL MEAL PLANNERS]

Product Requirements Document: New Feature

Feature name: [Meal Schedule]

Owner: [FitPrep project team: Yadan Tiano/ Racheal Winstead]

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1. PROBLEM

People working overnight or changing shifts need meal plans that are understandable across midnight and compatible with their waking hours and available breaks. Fitia's documented setup organizes meals into breakfast, lunch, dinner, and snacks, but the supplied documentation does not explain how to organize them around an overnight routine. Users may have to translate those categories into their own schedule, creating uncertainty about when to eat and which planning day contains a meal.

The problem hypothesis is that showing meals in the context of the user's schedule will reduce this planning effort. This has not yet been confirmed through user interviews or hands-on inspection of Fitia's full planner. Missing documentation is not evidence that Fitia lacks a capability.

1a. Background & Dependencies

Existing feature to recreate:** Personalized meal planning, including meal preferences and a daily meal-plan view. Fitia documents manual and automatic planning, food selection, variety, up to five meals per day, percentage calorie allocation, and simple-food or recipe suggestions.

- Documented entry path: Plan → profile photo → Quick Access → Meal Plan & Foods.

- Source: [How Can I Create My Meal Plan in

Fitia?](<https://fitia.app/help/articles/create-meal-plan-fitia/>), supplied by the project owner. [Fitia's meal-plan overview](<https://fitia.app/meal-plans/>) also describes meal swapping.

- Visual references: Two supplied onboarding screenshots show black backgrounds, white primary text, gray secondary text, yellow primary actions, rounded controls, and outlined nutrition summaries. They do not establish the planner's exact layout. Android status and volume overlays are not product components.

- Additional structural reference: The supplied Fitia meal-plan webpage text includes a product preview with "Today," a weekly date selector, calorie progress, protein/carbohydrate/fat progress, a Breakfast section, meal-level nutrition, and individual foods with quantities and calories. It also describes recipe suggestions with an "Add to Meal" action. This supports the information hierarchy to recreate, but does not verify live interactions or precise layout.

Promotional testimonials are not usability research or acceptance criteria. The preview's displayed fat totals differ between the daily summary and breakfast example, so its nutrition values must not be copied as a validated sample dataset.

- Validation dependency: Inspect actual meal-plan and preference screens before finalizing a faithful recreation or claiming a competitive gap. Confirm whether existing schedule controls already address this need.
- Prototype dependency: Use clearly identified sample meals and nutrition data. Live Fitia APIs, account access, and production recommendation infrastructure are not assumed available.
- Integration dependency: A production implementation must reuse existing dietary preferences, nutrition targets, meal generation, and logging behavior. Schedule presentation must not silently change calorie targets or duplicate food records.
- Business dependency: Confirm how this extension relates to Fitia's meal-planning entitlement before production release. The prototype demonstrates the experience without introducing a new payroll.

1b. Target Use Cases

1. As an overnight worker, I want to see meals before and after midnight in one coherent plan so that I can prepare for my entire waking period.
2. As a worker with limited breaks, I want to set meal times and labels that match my routine so that I can quickly identify the meal intended for my next break.
3. As a rotating-shift worker, I want to adjust one upcoming day without rebuilding my food preferences so that my plan stays useful when my hours change.
4. As a user following an individualized nutrition plan, I want schedule changes to preserve my nutrition targets so that organizing my meals does not override my established plan

1c. Current User Journey

The navigation and controls below come from the supplied help article. Pain points and workarounds are hypotheses for validation, not observed research findings.

1. User opens Fitia's "Plan" tab.
2. User taps their profile photo and selects "Meal Plan & Foods" from Quick Access.
3. User selects foods for breakfast, lunch, snacks, or dinner.
 - "Problem hypothesis:" A user waking in the afternoon must mentally map conventional meal names to their routine. The documentation does not explain alternative labels.
4. User chooses variety, up to five meals, calorie percentages, and simple foods or recipes.
 - "Problem hypothesis:" These documented choices do not explain where to enter overnight work hours or available meal breaks.
5. User saves preferences to create or update the plan.
 - Problem hypothesis: The documentation does not explain how a meal after midnight relates to the prior evening's plan. Actual calendar behavior remains unverified.

6. User may keep a separate note mapping planned meals to work breaks.
→ Problem hypothesis: Maintaining a separate schedule adds effort when shifts change.
Validate whether users actually use this workaround.

2. PROPOSED SOLUTION

Extend the existing meal planner with an optional “Meal Schedule” preference. Users can organize their planned meals around wake time, sleep time, and preferred eating times, including a waking period that crosses midnight.

How it works: From Meal Plan & Foods, the user opens Meal Schedule, enters their routine, and assigns times and optional custom labels to existing meal slots. A preview shows the meals in order with explicit calendar dates before the user saves. The Plan view then presents those same meals in the saved order while retaining existing nutrition information and meal actions.

2a. Value Proposition

Meal Schedule helps people working nontraditional hours prepare and follow a meal plan that matches their actual routine. Bringing schedule information into the existing planner reduces the need to translate meal categories or maintain a separate note, while preserving familiar food preferences and nutrition targets.

2b. Goals & Out-of-Scope

Goals

- Help overnight workers correctly identify the next planned meal and its date.
- Enable users to organize an overnight plan without assistance or a separate scheduling tool.
- Let users adjust upcoming meal times without rebuilding preferences or losing completed logs.
- Make the improvement feel like part of Fitia’s existing planner through consistent navigation, controls, and meal presentation.

Out-of-Scope

- Glucose prediction, diabetes treatment, fasting prescriptions, and automatic carbohydrate or calorie redistribution. This version improves scheduling; it does not claim to counteract the metabolic effects of night work.
- Employer-roster imports, automatic rotation detection, and complex recurring shift calendars. Users can edit an individual planning day in this version.
- New meal-generation infrastructure, wearable integrations, and a separate night-shift dashboard. The prototype uses sample content; production would depend on existing product services.

2c. Measurable Outcomes

Metric	How it's measured	Baseline	Target
[What you're tracking]	[How you'll measure it]	[Current state]	[Goal number or threshold]
Overnight-plan setup success	Moderated test: participant creates a plan containing a meal after midnight without facilitator assistance	Unmeasured; establish using the current planner before comparative testing	At least 8 of 10 target participants complete the improved flow without assistance
Meal-date comprehension	Ask each participant to identify the calendar date and planning day for a 1 a.m. meal	Unmeasured; establish in the same baseline study	At least 9 of 10 participants answer both correctly
Schedule-edit efficiency	Time from opening preferences to saving a one-day time change, among successful attempts	Unmeasured; time the comparable task in the current planner	Median of 60 seconds or less in the improved prototype

3. REQUIREMENTS

User Journey 1: Overnight worker setting up a usable meal schedule

Context: The user has meal preferences and nutrition goals and wants to organize meals around a routine that crosses midnight.

Sub-journey: [Finding and configuring Meal Schedule]

- **[P0]** User can access Meal Schedule within the existing Meal Plan & Foods preferences.
- **[P0]** User can keep the existing planning experience without enabling a custom schedule

- **[P0]** User can enter their usual wake and sleep times, including sleep occurring on the following calendar date.
- **[P0]** User can assign a local time to each selected meal slot while retaining the documented limit of five meals per planning day.
- **[P0]** User can see whether each meal time belongs to the starting date or the next date.
- **[P1]** User can add a custom display label such as “Before work” or “First break” while retaining the original meal category for food preferences.
- **[P1]** User can optionally enter work start and end times as reference information, without triggering dietary restrictions or automatic meal changes.

Sub-journey: [Previewing and saving]

- **[P0]** User can preview the chronological meal sequence, dates, and existing nutrition totals before saving.
- **[P0]** User can correct missing or conflicting time entries with an explanation that identifies the affected entry. Equal wake and sleep times require correction rather than an inferred 24-hour waking period.
- **[P0]** User can explicitly confirm an intentional meal outside the entered waking interval rather than having it silently moved or deleted.
- **[P0]** User can save schedule changes without changing food restrictions, calorie goals, macro targets, or meal calorie percentages.
- **[P0]** User can cancel changes and return to the prior saved schedule.
- **[P0]** User can retry a failed save without re-entering their choices and can distinguish an unsaved preview from a saved schedule.

Sub-journey: [Name of third step or flow]

- **[P0]** User can [critical action].
- **[P0]** User can [critical action].
- **[P1]** User can [important action].

User Journey 2: [Overnight worker following the plan across midnight]

Context: The user needs a clear answer to “Which meal is next?” even when the calendar date changes during their shift.

Sub-journey: [Reading the plan]

[P0] User can view an explicitly dated planning period containing all its meals, including next-date meals, within the existing Plan experience.

[P0] User can select a planning day using the familiar date-navigation structure and see calorie and protein/carbohydrate/fat progress above its meal sections. In custom schedule mode, the selected date is the period's starting date, and the complete date range is visible.

[P0] User can see each meal's name, scheduled time, calendar date, calories, and available macro information.

[P0] User can inspect the foods within a meal, including their quantities and calories, preserving the detail represented in Fitia's supplied website preview.

[P0] User can distinguish nutrition totals for the displayed planning period from calendar-day totals elsewhere in the product.

[P0] User can identify the next scheduled, incomplete meal without relying on color alone.

[P0] User can continue viewing the same plan after midnight without its unfinished meals disappearing solely because the date changed.

[P1] User can navigate to adjacent planning days and see their date ranges.

Sub-journey: [Using existing meal actions]

- **[P0]** User can open meal details and use the recreated meal-swap interaction from a scheduled meal.
- **[P0]** User can swap meal content while retaining its time, label, and planning-period assignment; updated nutrition information remains visible.
- **[P0]** User can record a meal once and see that same record associated with its planned slot, without generating a second nutrition-log entry.
- **[P0]** User can distinguish the scheduled time from the actual recorded eating time when they differ.
- **[P0]** User can access, read, and operate schedule controls with screen-reader labels, readable text scaling, and touch targets appropriate to the platform.

User Journey 3: [Worker adapting to a changed shift or day off]

Context: The user needs to change an upcoming routine without losing history or rebuilding their preferences.

- **[P0]** User can choose whether an edit applies to one selected planning day or their default routine for future, unstated planning days.
- **[P0]** User can edit times of unfinished meals in the current plan without moving or rewriting completed food logs.
- **[P0]** User can preview and resolve a scheduling conflict when edited periods overlap, rather than having a meal silently duplicated or reassigned.
- **[P0]** User can return to the standard planning view while retaining existing meals and their actual logged timestamps.
- **[P1]** User can reuse a saved schedule for another planning day.
- **[P2]** User can save named presets such as "Night shift" and "Day off."

Sub-journey: [Understanding the scope of guidance]

- **[PO]** User can understand that changing the schedule organizes meal timing and does not prescribe fasting or alter their nutrition targets.
- **[PO]** User can read the following contextual guidance during setup: “If you have diabetes, follow the meal plan and treatment instructions provided by your healthcare team. Consult them before changing meal timing, reducing carbohydrates, or fasting, especially if you use medication that can cause low blood sugar.”
- **[PO]** User can use the scheduling feature without disclosing a diabetes diagnosis. This version does not collect medical-condition information or provide condition-specific recommendations.

4. APPENDIX

Design Decisions

Extend the existing preference flow.** Place Meal Schedule inside Meal Plan & Foods and show its result in Plan. A separate shift-work tab would require users to learn and maintain an additional planning location.

Preserve the product’s visual language. Use the supplied screenshots to guide black surfaces, white and gray text, rounded shapes, and yellow primary actions. Confirm planner-specific components before final visual approval; do not assume onboarding is identical to the planner. Check contrast and readable control states independently of screenshot appearance.

Preserve the planner’s information hierarchy. The supplied website preview establishes a reference sequence: date navigation → calorie and macro progress → meal sections → foods and quantities. Place the planning-period date range near the date navigation and add the scheduled time and optional custom label to each existing meal heading. These are proposed placements, not observations of current Fitia controls. Keep the familiar food and nutrition details available rather than replacing the planner with a separate timeline.

Start with schedule clarity. The earlier nutrition discussion motivates the audience, but this release makes no glucose-lowering claim. Automated overnight food composition and calorie redistribution would require additional evidence, clinical review, and validation beyond this prototype.

Separate display grouping from food-log history. A planning period is identified by its starting date and explicitly spans at most 24 hours in the first version. For example, a period starting September 29 at 3 p.m. can include a meal on September 30 at 1 a.m. Each meal retains its calendar timestamp and a single planning-period association. Viewing it in a different grouping must not count it twice or rewrite its actual eating time.

****Make changing routines explicit.**** One-day overrides support basic rotating-shift needs. Full rotation automation is deferred to keep the feature testable and feasible for the project.

****Use a focused prototype.**** Demonstrate the existing planner, the Meal Schedule setup and preview, and the updated planner. Include one overnight example, one schedule edit, and one error state. Functional backend integration is not required to evaluate these flows.

Acceptance Scenario:

A user wakes at 3 p.m. on September 29, works from 7 p.m. to 7 a.m., and plans to sleep at 8 a.m. on September 30. They schedule meals at 5 p.m., 11 p.m., and 4 a.m. The preview and saved plan show all three meals together in chronological order, with the 4 a.m. meal explicitly dated September 30. Saving preserves their existing nutrition targets and calorie allocation. Moving the unfinished 4 a.m. meal to 3:30 a.m. changes only its planned time; recording it creates one food-log entry.

These times demonstrate interface behavior and are not a recommended dietary schedule.

Open Questions

| Does Fitia already support custom meal times, labels, or overnight grouping? If so, which remaining usability gap should this project address? | Project owner / UX research | Before approving the feature gap and final scope |

| What do the actual Plan, meal detail, and Meal Plan & Foods screens look like, including swap and logging controls? | Project owner / design | Before high-fidelity recreation |

| Do target users prefer a wake-based planning period or a configurable daily boundary? | UX research | Before interaction design is finalized |

| How should time-zone travel and daylight-saving changes affect scheduled times and date grouping? The initial prototype assumes one fixed local time zone. | Product / engineering | Before production implementation |

| How does period-based presentation interact with existing calendar-day reports and any shared plans? | Product / engineering | Before production data-model approval |

| Should the improvement inherit existing meal-planning entitlement? | Product owner | Before production release |

| What are the baseline task results, and are the proposed usability targets realistic? | UX research | Before comparative prototype testing |

Other links

- [Fitia meal-plan creation instructions](https://fitia.app/help/articles/create-meal-plan-fitia/)
- [Fitia automatic-plan instructions](https://fitia.app/help/articles/activate-automatic-meal-plan-fitia/)
- [Fitia meal-plan overview](https://fitia.app/meal-plans/)
- [NIDDK: Healthy Living with Diabetes](https://www.niddk.nih.gov/health-information/diabetes/overview/healthy-living-with-diabetes), supporting the caution about changing meal timing with medications that can cause low blood sugar.
- Visual reference: user-supplied “Track your meals in seconds” onboarding screenshots, including the grilled-chicken nutrition example. These are design references, not evidence of the current planner’s behavior.
- Structural reference: user-supplied “A Science-Based Meal Plan That Actually Works” webpage text, including the Today/date-navigation preview, nutrition progress, Breakfast food list, and description of recipe suggestions. This is marketing-page evidence, not a live-app usability test.
- Source template: user-supplied “[FEATURE NAME] Product Requirements Document: New Feature.” Its section structure is retained here; directions and example placeholders have been removed.